

AYUSH KUMAR BISWAL

Applied AI / AI Product Engineer | B.Tech Computer Science & Engineering, Class of 2026

+91 7735172340 | ayushkumarbiswal@gmail.com | Odisha, India

linkedin.com/in/ayush-kumar-biswal | github.com/ayushkumar1316 | ayush-portfolio-sable-nu.vercel.app

PROFESSIONAL SUMMARY

Final-year Computer Science student (2026) building AI agent systems and LLM-powered full-stack applications. Built a working autonomous task-worker prototype that interprets natural-language goals, plans action sequences, selects tools, executes them, retries on failure and verifies completion before reporting success. Ships production-style LLM features using structured outputs, schema validation and API integration, and uses AI coding tools (Claude Code, Cursor) daily to move faster and debug unfamiliar codebases quickly.

TECHNICAL SKILLS

AI & Agents: LLM Integration, Prompt Engineering, Agentic Task-Execution Loops, Tool Selection & Execution, Structured Outputs / Function Calling, Schema Validation (Zod), Retrieval-Augmented Generation (RAG), Output Verification, Failure Handling & Retries, Evaluation

Languages: JavaScript, Python, Java, SQL, HTML5, CSS3

Frontend: React.js, Responsive Dashboard & UI Design

Backend & Data: Node.js, Express.js, Flask, REST APIs, MongoDB, MySQL, Firebase

Machine Learning: XGBoost, Random Forest, Imputation, Scaling, Feature Selection, Class Balancing

Tools & Workflow: Git, GitHub, Postman, VS Code, Figma, AI-assisted coding (Claude Code, Cursor)

Concepts: APIs & External Integrations, Browser Automation, Human-in-the-Loop Approval Flows, Debugging & Root-Cause Analysis

PROJECTS

Codex Canvas — AI-Powered Visual Development Platform — Structured, Verified AI Edits for Web Interfaces | React.js, JavaScript, AI, Structured Outputs, Zod

- Built an AI-assisted development platform that creates and modifies web interfaces through constrained structured edit operations rather than unrestricted free-form model output.
- Implemented 8+ API-backed editing operations with Zod schema validation, business-rule checks and component-registry verification executed before any AI change was applied.
- Reached 89% successful edit-operation completion across 50+ validation rules by constraining tool selection to a validated action space, so malformed or unsafe model output fails as a catchable validation error instead of corrupting the UI.
- Separated reusable components from AI-driven edit logic to keep the frontend architecture maintainable and debuggable across repeated AI edits.

FlowMind — AI Execution Coach — LLM Reasoning Over Deadlines, Progress and Risk | React.js, JavaScript, AI/LLM Integration

- Built an LLM-based execution coach that analyses deadlines, task progress and execution risk to generate actionable plans across 200+ tasks.
- Delivered a responsive dashboard with confidence scoring, risk analysis, deadline tracking, focus planning and recovery planning, reaching 85% accuracy in confidence scoring against real task-progress data.
- Normalised deadline, progress and risk inputs into a single-source-of-truth data structure before LLM analysis, and used structured LLM outputs so the model returned consistent, machine-readable planning results instead of free-text that could not be verified.

AQI Predictor — Machine Learning Web Application — ML Backend Connected to a Working App Interface | Python, Flask, XGBoost, Random Forest, REST API

- Built a web-based AQI prediction application on Indian air-quality data, integrating XGBoost and Random Forest models behind a Flask backend to reach 92% prediction accuracy.
- Cut average prediction response time to 2.1s by exposing predictions through REST APIs and wiring the ML backend to the application interface end to end.
- Built a preprocessing pipeline of imputation, scaling, feature selection and class balancing to keep the models stable on incomplete, imbalanced air-quality records.

AgriPlus — Direct Market Access for Farmers — Android App for Farmers, Markets and Experts | Java, XML, Firebase, MySQL, Android

- Built an Android application connecting farmers directly to market opportunities and agricultural resources, with modules for farmer-market interaction, educational content and expert assistance backed by Firebase and MySQL.

EXPERIENCE

Independent Full Stack Developer | Remote | 2025 – Present

- Design, build and debug full-stack web applications end to end with React, JavaScript, Node.js, Express, Python and databases, taking projects from requirements to a working deployed application independently.
- Develop REST APIs and integrate frontend and backend services, resolving data-model and integration failures through systematic root-cause debugging rather than guesswork.
- Manage projects and version history with Git/GitHub, and use AI coding tools to explore unfamiliar codebases, read new frameworks quickly and accelerate iteration on unfamiliar technology.

EDUCATION

B.Tech, Computer Science & Engineering — GIET University, Gunupur, Odisha | 2022 – 2026 (Expected)

ADDITIONAL

- Smart India Hackathon Finalist — selected among the top 10 teams out of 2,000+ entries.
- TCS CodeVita — national rank 5,771.